Research proposal

Designing “BATTLE4GMP”: A GMP Educational Game for Pharmaceutical Sciences using Mechanics-Dynamic-Aesthetic (MDA)Framework

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1. Introduction

Learning pharmaceutical science can be challenging especially when most of the information is theory based. This leads to confusion on pharmaceutical students as learning theory and applying it in real life situation can be a different and demanding. The old ways of learning are insightful but can get boring over time and this may lead students to forgetting crucial information that can lead them into making mistakes that can jeopardise production of quality pharmaceutical products or even safety on the patients. Game based learning is a modern approach that can be used for enhancing students learning, to help them apply the knowledge they acquired in class through real life scenario questions that require critical thinking. **MDA FRAMEWORK**

2. Background

Learning pharmaceutical science falls short in preparing the students for the high demanding working environment. *(Oestreich, 2022)* Due to the evolving technology and demand for improvement of pharmaceutical skills, there is a growing need for pharmaceutical professionals who possess skills like good decision making, critical thinking and have good interprofessional relationships. Current teaching practises focus on traditional ways of learning through theory and are not practically based so this gives student the foundational knowledge but fails to prepare them to use it in the work environment. *(Hope et.al,2023)* Game based learning will help bridge this gap helping pharmaceutical students and personnel to integrate and apply their knowledge in high pressure situations. Literature articles and studies show positive improvement in students who were learning using pharmaceutical games like the escape room proving the need for including innovation.

3. Problem statement

Good manufacturing practice is an important component of pharmaceutical education and practice, as it improves principles and standards necessary to ensure product quality, safety and consistency. However, GMP is still aa complex concept for many people including pharmacy students and industry personnel. This creates a challenge for understanding GMP and applying its core principles in real life situations. Pharmacy students and industry personnel need to have a continuous understanding of GMP principles and how to apply them in the professional space. Currently there are limited interactive learning activities such as educational games that can help students engage with GMP concepts in a practical setting.

The lack of GMP focused learning activities makes it difficult for students to bridge theory part of the GMP with real life practices. This gab highlights the need for an innovative educational game-based learning programme that allows students to engage with GMP content in a fun but education way. Battle4gmp was developed as a fun and serious educational game that incorporates gaming elements with educational principles offering a fun academic way to learn GMP principles. The proposal therefore seeks to design a game based on GMP to support pharmacy students and pharmaceutical industry personnel in overcoming learning barriers

4. Research question

How can the MDA (mechanics, dynamics, and Aesthetics) framework be integrated into the design of a GMP based educational game called ‘Battle4gmp’ to enhance the learning and engagement among pharmacy students and pharmaceutical industry personnel?

5. Research aim and objectives

5.1. Aim

The aim of developing BATTLE4GMP is to teach and reinforce key Good Manufacturing Practice (GMP) principles, Data integrity (ALCOA), Personnel requirements, and Sterility Assurance. The game will be structured using the Mechanics-Dynamics-Aesthetics (MDA) Framework to create an interactive learning space. Within the game, players will apply GMP concepts through problem solving, recall, recognition, decision-making, and receive immediate feedback. This approach is supported by evidence from Kanaan et al. (2023), who found that game-based learning in pharmacy education positively impacts engagement, motivation, understanding, confidence, critical thinking, and academic performance. It is important to note that the focus of this project is on the systematic design and development of the educational game, not the effectiveness. This aligns with GBL proposal guidelines, where the MDA Framework serves as the research methodology and finished game is the primary research output.

5.2. Objectives

The objectives of the game are to:

- Review literature regarding current challenges in GMP or pharmaceutical education, existing teaching approaches, serious game design, evidence-based principles of game-based learning for the development of BATTLE4GMP
- Identify and formally define the intended learning outcomes for each of the three pharmaceuticals domains: Data integrity (ALCOA), Personnel (Annex 1 cleanroom practices, gowning, and grade classifications), and sterility assurance (contamination control and sterility vocabulary)
- Produce a fully playable, interactive prototype of the BATTLE4GMP game, including all three levels, the Grand reward (interactive book), scoreboard tracking, and the complete clue-token resource management system.
- Design engaging aesthetic experience that promotes curiosity, challenge, discovery, accomplishment, and confidence. These experiences are intended to make GMP learning more interactive and enjoyable while supporting sustained engagement with educational content.
- Develop game dynamics that encourage problem-solving, strategic thinking, decision-making, and error correction as players interact with the game mechanics, this will encourage players to actively process GMP information rather than simply receive information passively.
- Provide immediate and meaningful feedback during gameplay to help players recognise correct and incorrect responses and encourage continued engagement. BATTLE4GMP will therefore incorporate features such as word level validation, colour-coded feedback and scoring to make players progress visible and understandable
- Ensure that the game provide opportunities for repeated, risk-free engagement with GMP concepts, allowing players to make mistakes, receive feedback and attempt challenges without consequence associated with real pharmaceutical manufacturing environments. Game-based learning has been identified as particularly useful in health-professions education because it can provide opportunities for real-world application and risk-free decision-making.

6. Methodology

6.1. Educational Analysis

BATTLE4GMP is a game designed to meet the learning needs of pharmaceutical students and personnel. The game will focus on topics like data integrity, Alcoa principles and different types of personnel and their roles in the pharmaceutical industry.

The learning outcome for the game is to help pharmaceutical student and the pharmaceutical industry personnel about the different principles of ALCOA+, know the explanation of the terminologies and how to apply them.

- Data integrity: The definition of data integrity, regulatory requirements that are needed to be complied with. How data is validated and the different data integrity risks associate with paper based, hybrid or computer stored data.
- Alcoa principles: Understand the different types of ALCOA+ principles, what they mean and how they should be applied in the pharmaceutical industry
- Personnel: Learn about different personnel in the pharmaceutical environment and their roles. The restriction on the movement of personnel in different grade to minimise the risk of cross contamination. How maintenance is done in each grade and the control measures and principles that are put into place.

This will assist in installing the necessary needed knowledge in second, third and fourth year students who are doing pharmaceutical science and personnel working in the pharmaceutical industry. Battle4GMP does not have an educational requirement so everyone can play it, but it was designed specifically for pharmaceutical students and pharmaceutical industry personnel.

6.2. Mechanic

The process of translating educational objectives into a game mechanics can be guided by established frameworks within serious game design literature. Arnab et al. 20215 introduced the learning mechanics game mechanics (LM-GM) model, which is based on the principle that serious game design involves converting learning goals and practises into gameplay, mechanics that serve an instructional purpose in addition to providing entertainment. The model incorporates predefined learning and game mechanics derived from game studies and learning theories, enabling designers to construct an LM-GM map that illustrates the relationship between educational objectives and game-based features. This model is relevant to BATTLE4GMP because the game aims to prepare pharmacy students to apply GMP principles in pharmaceutical manufacturing contexts, were serious inappropriate decisions or error may have serious implications.

For BATTLE4GMP, this systematic mapping guided the design of all three game levels. In levels 1 data integrity, the objectional objective of recalling ALCOA and principles was translated into a crossword mechanic where players must type complete word to fill blanks, receiving immediate feedback through colour coded validation. Level 2personnel uses a drag and match mechanic that requires players to connect personnel related items with their correct definitions, directly supporting the objective of distinguishing between personnel requirement under annex 1. Level 3 sterility employs a word search mechanic to reinforce sterility vocabulary recognition, allowing players to identify hidden terms within a letter grid. Each level also includes incorporates a clue-token system and a +3 bonus for perfect performance, which supports the dynamics of strategic decision-making and error correction. This alignment between learning objectives and game mechanics ensures that gameplay is not merely entertaining but purposeful designed to reinforce the intended pharmaceutical knowledge.

6.3. Dynamics

The dynamics of BATTLE4GMP describe how players will interact with the game mechanics to create a meaningful learning experience. When a player engages with mechanics, a series of actions, feedback loops, and decision-making process will unfold, reinforcing GMP principle in a risk-free environment

Level 1: Data integrity (Crossword)

When a player interacts with the crossword grid, a dynamic cycle of recall, validation and correction is initiated

- Actions & Feedback: As the player types a letter into a blank space, the game provides immediate, colour-coded feedback. A correct letter will lock into space (e.g. turn green), while an incorrect one will be rejected or highlighted (e.g. turn red), prompting the player to reconsider their answer.
- Problem-solving & resource management: if a player struggles to recall a term, they can choose to use a “clue-token,” which will reveal a letter or hint. This creates a strategic dynamic, forcing the player to rely on their existing knowledge and use their limited resources, thereby promoting active retrieval and problem-solving rather than passive guessing. The pressure to complete the grid correctly to earn +3 bonus encourages careful consideration and error correction.

Level 2: Personnel (Drag and Match)

The dynamics in this level centre on application, analysis, and correction of misconceptions.

- Application and analysis: The player drags a term from a list and drops it onto a matching definition. The act of matching is an active analysis, requiring the player to distinguish between similar concept such as differing gowning requirement for Grade B and Grade D cleanrooms
- Immediate clarification: upon dropping a term, the game will instantly validate the match. A successful match will trigger a positive reinforcement or a brief explanatory pop-up, while an incorrect match will be rejected and term will snap back to its original position. This dynamic forces the player to mentally process the definitions and relationships between concepts, correcting misunderstanding immediately rather than reinforcing them
- strategic thinking: The strategic use of clue-tokens continues here allowing players to reveal correct match if they are uncertain.

Level 3: sterility (word search)

The dynamics of this level shift from simple vocabulary recognition to higher-order applications, requiring players to connect definitions or scenarios to their corresponding sterility terms

- Definition-driven search: unlike a traditional word search where players look for words in isolation, level 3 presents players with a definition or a brief scenario at the bottom of the screen. The player must first read and comprehend the description, then actively search the letter grid for the sterility term that matches it. This dynamic promotes deeper cognitive processing, as the player must interpret the meaning of the term before locating it, rather than simply recognising it as a string of letters
- Active recall & application: when a player clicks on a scenario, it becomes highlighted, focusing their attention on the specific concept they need to find. The player then scans the grid, selecting letters to form the matching word. This dynamic transforms the experience from passive recognition into active recall and application, reinforcing the practical meaning of sterility vocabulary
- Immediate validation and reinforcement: once the player selects the correct sequence of letters corresponding to the highlighted scenario, the word validated and locked in place (e.g. highlighted in green). A brief definition or contextual explanation will appear to reinforce the learning. If the player selects an incorrect sequence, the game provides feedback (e.g. the letters are rejected or the player is prompted to try again), encourage them to re-evaluate their understanding of the scenario and search again.
- Task completion & progression: The dynamic of completing one matched pair unlocks the scenario, creating a clear sense of progression. this step-by-step approach ensures that players engage with each sterility concept individually, building confidence and reinforcing vocabulary in a structured manner. The 3+ bonus for perfect performance incentivises careful reading and accurate searching, promoting thorough engagement with the material.

Cross-Level dynamics:

- Progression & achievement: the game unlocks sequentially (Level 1–>2–>3), creating a dynamic of progression. Each successful level completion is met with immediate positive feedback (score accumulation, unlocking the next level), driving player and providing a sense of achievement)
- Risk-free failure: crucially, a player’s inability to solve a puzzle does not end the game. The dynamic is one of iteration. If a player completes a crossword with an error, they can submit after correction. This encourages risk-taking, experimentation, and learning from mistakes without a fear of real-world consequences. The immediate feedback loops at every level allow players to continuously self-assess and build confidence in their knowledge.
- Knowledge integration: the progression through the three distinct domains (data integrity, personnel, sterility) creates a dynamic where the player builds a holistic understanding of GMP, seeing how each pillar is essential for quality pharmaceutical production)

6.4. Aesthetics

Describe the intended player experience and explain why these experiences support learning

6.5. Interactive design

To refine BATTLE4GMP we will use:

- PEER REVIEW- where our game will be played by our peers and lecturers so they can give us feedback required to improve the game's design and fix whatever issues are there. Their feedback will be in a form of survey or reflection that is compiled and read ethically without any bias and used to perfect the design of the game
- Lecturer feedback will also be considered where we will showcase our proposal and get feedback about where our game lacks and how we can improve it so our target audience can not only play the game but also learn and acquire knowledge from it simultaneously as this is a gamed based on learning
- INTERNAL TESTING DESIGN-the designers will play the game multiple times to see the errors or any gaps that need to be fixed in the game
- Repeated prototype design- The designers of the game will repeatedly reassess the game to ensure its design is at its optimal level until all errors are corrected

6.6. Final product

The competed BATTLE4GMP game will consist of three levels of which each will consist of a different game designed to reinforce a specific GMP concept. The game will use interactive activities to encourage learning and engagement while applying GMP concepts.

- Level 1 is a crossword puzzle game that will focus on the concept of data integrity with emphasis on the ALCOA+ principles. Players will be required to complete a crossword puzzle using key terms and concepts related to data integrity. this level aims to reinforce understanding of the ALCOA+ principles and the importance of maintaining data integrity in the pharmaceutical industry.
- Level 2 is a drag the words game that will focus personnel, and the roles and responsibilities personnel has within the pharmaceutical industry. Players will be required to drag and drop words to match a specific personnel member to the roles corresponding to their responsibilities such as identifying the role a warehouse pharmacist plays in the industry. This level aims to improve players understanding of the roles and responsibilities played by personnel and how they contribute to maintaining the overall GMP standard.
- Level 3 is a word search game that focuses on sterility and its relation to the GMP concepts, it is different from the typical word search in that definitions will be provided as clues, requiring players to identify the corresponding terms withing the word search, in doing so this level aims to encourage players to apply their knowledge of sterility concepts and the key terminologies associated with maintaining sterility and preventing contamination the pharmaceutical industry.

The battle4gmp prototype will incorporate components to support learning through gameplay experience.

Component	Description	Purpose
Digital interface	Interface that players can use to access the game, move through level and intact with the game	to provide access and user-friendly platform

instruction manual	Provides instructions on how to play BATTLE4GMP, which includes rules	To help guide the players and ensure understanding
Learning outcomes	Each level will have specific learning outcomes linked to the specific GMP concept	To ensure that the intended objectives are met
interactive activities	three activities which include a crossword puzzle, drag the words and word search	Reinforces GMP concepts through interactive activities
Feedback	Players will receive immediate feedback after submitting their responses	To help players identify their mistake and where they went wrong
visuals	The game will include graphics, icons and background elements related to GMP	To make the game fun and appealing to engage with
level completion	Players will be able to progress though the levels by completing activities	Provides sense of achievements and encourages players to complete all levels

7. Ethical considerations

This research focuses on design and development of the BATTLE4GMP game, therefore the ethical issues to consider are related to the design and development of the game.

- Copyright and intellectual property: copyright and intellectual property will be strictly considered during the design and development of BATTLE4GMP. Any images, texts, icons, and graphics used will be selected and used in accordance with the copyrights permits and requirements.
- Proper referencing and acknowledgement of sources: all information and materials used in developing the game will be obtained from credible, peer-reviewed sources. Educational content used to comply Battle4GMP content, questions and learning outcomes will be properly acknowledged and referenced to maintain academic integrity.
- use of Artificial Intelligence: artificial intelligence tools will be used as supporting sources for tasks such as logo designs and assistance with brainstorming ideas. Any AI generated information will be critically reviewed before being incorporated into the final game. Researchers will remain responsible for the final content of the game and will disclose any AI used.
- Accuracy of educational content: educational content that will be used in BATTLE4GMP will be reviewed to ensure its accuracy and relevance to the GMP principles. All the content used to comply the questions will be revied against the official GMP guidelines to maintain integrity and reliability to the objectives.
- Acknowledgement of contributors: individuals who are contribute to the design and development of the game will be appropriately acknowledged for their input.

8. Limitations

Several limitations are acknowledged for this research project. Firstly, this study focuses on the systematic design of an educational game using the MDA framework, rather than evaluating its impact on learning outcomes. As such, the project does not include experimental testing, pre- and post-assessment, or comparative analysis to determine whether the game improves knowledge retention or clinical reasoning skills. The game's educational effectiveness is therefore not empirically validated within the scope of this research. Secondly the game is designed specifically to teach Good Manufacturing Practice (GMP) principles, with a focus on three levels: Data integrity (ALCOA+), Personnel requirements, and Sterility Assurance. As a result, the game does not cover other important pharmaceutical topics such as quality management system, Premises and Equipment, Deviations, and root cause analysis, which may limit its broader applicability within the pharmacy curriculum.

The game is designed specifically for pharmaceutical students and industry personnel in South Africa and the findings from the design process may not be generalised to other educational settings, or student populations without significant adaption. Moreover, the game incorporates digital elements, so students without access to appropriate devices or reliable internet connectivity may face barriers to participation, which is particularly relevant in resource-limited educational settings. The game is also designed to be completed within a specific timeframe, and longer gameplay session may not be feasible within the current curriculum structure. This may limit the depth of content that can be covered or complexity of the game mechanics that can be included.

The game will not be implemented or tested in an actual educational setting during this project, meaning there is no data on learner engagement, user experience, or practical feasibility of integrating the game into the existing pharmacy curriculum. This would require future research involving student participants and real-world teaching environment. Finally, the game prototype may require ongoing maintenance, updates, or replacement of physical components over time. Without institutional support or funding, the long-term sustainability of the game as a teaching tool may be uncertain.

These limitations are not weakness of the study but rather boundaries define its scope. Future studies could extend this work by conducting effectiveness testing with pharmacy students, implementing the game in classroom settings, refining the prototype based on user feedback and expanding the game to include additional GMP topics or advanced levels.

9. Timeline

Phases	Mar 2026	Apr-May 2026	Jun-Jul 2026	Aug 2026	Sep 2026	Oct 2026	Nov 2026
Literature review							
Background							

study and brainstorming							
Game design (MDA framework)							
Development of the game							
Gaming presentation and voting							
Hand in first Proposal draft							
Abstract writing workshop							
Hand in final draft abstracts							
Final write-up							
Poster compilation and presentation							
hand in final abstracts							
Hand in final poster submission							
Research symposium							
Hand in final write up submissions							
Hand in research portfolios							

10. Badget

Item	Estimated cost	Notes
Wi-Fi/internet	R 0.00	Used nmu Edu roam
H5p	R 0.00	free trial

AI	R 100	

11. References

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12. Appendices

12.1. Appendix A: Initial Game concepts

12.2 Appendix B: BATTLE4GMP logo development

Figure B1: final BATTLE4GMP logo

Figure B2: Initial logo ideas

12.3 Appendix C: BATTLE4GMP Storyboard